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List of all the I/O needed for your project.

1. Camera to Jetson connected serially via USB 2.0 (Capture Node)
2. Jetson to LabVIEW (Host Node): sends the captured, downsampled, raw pixel data from the image capture node to the central host via UART.
3. LabVIEW to the ESP32 Feather V2 (Display Node): Grabs fresh, newly filtered pixel data from the host computer and brings it into the ESP32 to pre-process it for the LED matrix display via UART.
4. Potentiometer to ESP32 (Analog input): a GPIO pin reads the analog/digital state of the pot and will trigger a visual filter change on the Host and Display nodes.
5. Could use I2C i.e. SDA/SCL if we decided to change the pot to a I2C rotary encoder or something similar.